

TITAN A16 - CAD MODEL SPECIFICATION

Parametric Geometry for SolidWorks, Fusion 360, or Onshape

Export Format: STEP AP214 (ISO 10303)

Units: Millimeters

Coordinate System: Right-hand, Z-up

BUILD SEQUENCE FOR CAD SOFTWARE

STEP 1: BASE CYLINDER

Command: Revolve / Cylinder

Parameters:

...

Datum: XY plane at Z=0

Diameter: Ø30.00 mm

Height: 10.00 mm (along +Z axis)

Center: X=0, Y=0, Z=0 (origin at bottom center)

...

Result: Solid cylinder (30mm dia × 10mm tall)

STEP 2: CENTRAL BORE

Command: Cut-Extrude / Hole

Parameters:

...

Sketch Plane: XY plane at Z=0

Circle Center: X=0, Y=0

Circle Diameter: Ø3.03 mm

Extrusion Direction: +Z

Depth: 10.00 mm (through entire height)

...

Result: Hollow cylinder with central bore

STEP 3: TOP LATTICE NODE POCKETS (12x)

****Command:**** Circular Pattern of Cut-Extrude

****Base Pocket Sketch:****

...

Sketch Plane: XY plane at Z=10.0 (top face)

Circle Center: X=5.00, Y=0.00

Circle Diameter: Ø2.00 mm

Extrusion Direction: -Z

Depth: 1.00 mm

...

****Pattern Parameters:****

...

Pattern Type: Circular

Center Point: X=0, Y=0, Z=10.0

Number of Instances: 12

Angular Spacing: 30° (360°/12)

...

****Verification - Node Coordinates (for manual placement if pattern fails):****

Node	Angle	X	Y	Z-Start	Z-End
1	0°	+5.000	0.000	10.0	9.0
2	30°	+4.330	+2.500	10.0	9.0
3	60°	+2.500	+4.330	10.0	9.0
4	90°	0.000	+5.000	10.0	9.0
5	120°	-2.500	+4.330	10.0	9.0
6	150°	-4.330	+2.500	10.0	9.0
7	180°	-5.000	0.000	10.0	9.0
8	210°	-4.330	-2.500	10.0	9.0
9	240°	-2.500	-4.330	10.0	9.0
10	270°	0.000	-5.000	10.0	9.0
11	300°	+2.500	-4.330	10.0	9.0
12	330°	+4.330	-2.500	10.0	9.0

STEP 4: TRIPLE HELIX THREADS

****Method A: Helical Cut (Preferred for CNC visualization)****

****Helix Path Parameters:****

...

Start Point: X=12.5, Y=0, Z=0

Helix Axis: Z-axis (vertical)

Pitch: 3.333 mm per revolution

Revolutions: 3.0 (total rise: 10mm)

Direction: Right-hand (counterclockwise looking down)

...

****Thread Profile (Sketch on helix plane):****

...

Profile Shape: Rectangular groove

Width: 1.0 mm (tangent to helix)

Depth: 0.5 mm (radial inward)

...

****Generate 3 Helices:****

1. ****Helix 1:**** Start angle = 0°
2. ****Helix 2:**** Start angle = 120° (offset from Helix 1)
3. ****Helix 3:**** Start angle = 240° (offset from Helix 1)

****Each helix:****

- Subtract thread profile swept along helix path
- Results in 3 interwoven grooves

****Method B: Equation-Driven Curve (Advanced)****

For software that supports parametric curves (Fusion 360, Onshape):

****Helix Equation (Cylindrical Coordinates):****

...

$r(t) = 12.5 \text{ mm}$ (constant radius)

$\theta(t) = t \times 360^\circ \times 3$ (3 full revolutions)

$z(t) = t \times 10.0 \text{ mm}$

where t ranges from 0 to 1

...

****Convert to Cartesian for CAD input:****

...

$x(t) = 12.5 \times \cos(t \times 1080^\circ)$

$y(t) = 12.5 \times \sin(t \times 1080^\circ)$

$$z(t) = t \times 10.0$$

Parameter t: 0 → 1 (use 360 points for smooth curve)

...

STEP 5: AXIAL LAYER BOUNDARIES (OPTIONAL VISUAL)

****These are composition zones, not physical cuts.**** For visualization only:

****Create 5 horizontal planes:****

...

Z = 0.00 mm (bottom)

Z = 2.00 mm (L1/L2 interface)

Z = 4.00 mm (L2/L3 interface)

Z = 6.00 mm (L3/L4 interface)

Z = 8.00 mm (L4/L5 interface)

Z = 10.00 mm (top)

...

****Split body at planes**** (creates 5 solid regions)

****Assign material appearances:****

- L1, L3, L5: Titanium (silver-gray)

- L2, L4: Gold (yellow-gold)

****Note:**** This is for visualization only. The actual part is monolithic, not stacked layers.

VERIFICATION CHECKLIST

After completing CAD model, verify:

****Dimensions:****

- [] OD = Ø30.00 mm exactly

- [] Height = 10.00 mm exactly

- [] Central bore = Ø3.03 mm exactly

- [] Bore is concentric with OD (check in section view)

****Features:****

- [] 12 node pockets present on top face

- [] Each node pocket: Ø2.00 mm × 1.00 mm deep

- [] Node pockets positioned at 5.00 mm radius
- [] Triple helix threads visible on OD surface
- [] Helix threads do not intersect node pockets

****Mass Properties (Approximate for Ti-dominant alloy):****

...

Expected Mass: 25-35 grams (depends on Ta/Ti/Au ratio)

Volume: ~6,500 mm³ (accounting for bore and pockets)

Center of Mass: X≈0, Y≈0, Z≈5.0 mm (axial center)

...

EXPORT SETTINGS

For Machine Shop (STEP File):

...

Format: STEP AP214

Units: Millimeters

Tolerance: 0.001 mm

Surfaces: All (include hidden geometry)

Assembly: Single part (not assembly)

Coordinate System: Match WCS (Z-up, origin at bottom center)

...

****File Name:**** `TITAN\_A16\_v2.1\_MASTER.step`

For FEA Import:

...

Format: Parasolid (.x_t) or IGES

Mesh Resolution: Native CAD (not faceted)

Include: All solid bodies

...

PARAMETRIC TABLE (FOR CAD VARIABLES)

If your CAD software supports design tables, use these variable names:

Parameter Name	Value	Units	Description
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`OD`	30.00	mm	Outer diameter
`HEIGHT`	10.00	mm	Total height
`BORE\_DIA`	3.03	mm	Central bore diameter
`NODE\_RAD`	5.00	mm	Node pattern radius
`NODE\_DIA`	2.00	mm	Node pocket diameter
`NODE\_DEPTH`	1.00	mm	Node pocket depth
`NODE\_COUNT`	12	—	Number of nodes
`HELIX\_RAD`	12.5	mm	Helix path radius
`HELIX\_PITCH`	3.333	mm	Pitch per revolution
`HELIX\_TURNS`	3.0	—	Number of revolutions
`HELIX\_STARTS`	3	—	Number of helix threads
`THREAD\_WIDTH`	1.0	mm	Helix groove width
`THREAD\_DEPTH`	0.5	mm	Helix groove depth

****Edit these values to regenerate geometry**** (useful for design variations)

RENDERING SUGGESTIONS

For presentation or grant proposal visuals:

****Material Appearances:****

- ****Titanium zones:**** Brushed aluminum (matte, gray)
- ****Gold zones:**** Polished gold (reflective, warm)
- ****Bore interior:**** Darker gray (precision-machined)

****Lighting:****

- 3-point studio lighting
- HDRI environment (optional)
- Show reflection on gold interfaces

****Camera Angles:****

1. Isometric view (show overall geometry)
2. Top view (show 12-node pattern clearly)
3. Side view (show helix threads)
4. Section view (cut at Y=0, show bore and internal layers)
5. Close-up (detail of helix intersection with node pocket)

ALTERNATIVE: ONLINE CAD (ONSHAPE) INSTRUCTIONS

If using Onshape (free for public documents):

1. **Create new document:** "TITAN A16"
2. **Create Part Studio:** "Main Geometry"
3. **Follow STEP 1-5 above** using:
 - `Cylinder` feature for base
 - `Hole` feature for bore
 - `Circular Pattern` for nodes
 - `Helix` feature for threads (under Insert > Curve > Helix)
 - `Sweep` to create thread profile along helix
4. **Export:**
 - Right-click part → Export
 - Format: STEP
 - Units: Millimeter
 - Download
5. **Share link** with machine shop or for grant documentation

SIMPLIFIED GEOMETRY (IF CAD IS TOO COMPLEX)

For initial quotes or FEA proof-of-concept, you can omit:

- Triple helix threads (simplify to smooth OD cylinder)
- Top node pockets (simplify to flat top face)

Keep only:

- Base cylinder $\text{Ø}30 \times 10 \text{ mm}$
- Central bore $\text{Ø}3.03 \text{ mm}$
- Axial layer split (for FEA material assignment)

This reduces modeling complexity while preserving core geometry for analysis.

APPENDIX: PYTHON SCRIPT FOR HELIX POINT GENERATION

For CAD software that accepts point clouds or spline imports:

```
```python
import numpy as np
import csv

Parameters
```

```

radius = 12.5 # mm
height = 10.0 # mm
turns = 3.0
num_points = 360 # points per helix
num_helices = 3

Generate helix coordinates
def generate_helix(start_angle_deg, output_file):
 t = np.linspace(0, 1, num_points)
 theta = np.deg2rad(start_angle_deg) + t * turns * 2 * np.pi

 x = radius * np.cos(theta)
 y = radius * np.sin(theta)
 z = t * height

 # Write to CSV
 with open(output_file, 'w', newline='') as f:
 writer = csv.writer(f)
 writer.writerow(['X', 'Y', 'Z'])
 for i in range(len(x)):
 writer.writerow([f'{x[i]:.4f}', f'{y[i]:.4f}', f'{z[i]:.4f}'])

 print(f"Generated {output_file} with {num_points} points")

Generate 3 helix paths
generate_helix(0, 'helix_1_0deg.csv')
generate_helix(120, 'helix_2_120deg.csv')
generate_helix(240, 'helix_3_240deg.csv')

print("Import these CSV files into CAD as XYZ point curves")
'''

Usage:
1. Run script (outputs 3 CSV files)
2. Import each CSV into CAD as curve/spline
3. Use curves as sweep paths for thread profile

QUESTIONS?

CAD Support Contact:
Matthew James Campbell
[YOUR EMAIL]

```



For machine shop CAD format requests, we can provide:

- STEP AP214 (universal)
- Parasolid (.x\_t for Siemens NX)
- IGES (legacy systems)
- Native SolidWorks (.sldprt) if needed

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**\*\*END OF CAD SPECIFICATION\*\***

\*Use this document alongside G-code and RFQ to ensure geometric consistency across all deliverables.\*